🏃 Strava Fitness Data Analysis Report

1️⃣ Project Overview

This project analyzes user fitness data from a wearable fitness tracking application similar to **Strava**. The dataset contains activity, sleep, heart rate, and weight information recorded from multiple users.

The purpose of this analysis is to identify **user activity patterns, health indicators, and engagement behaviour** to generate insights that could help improve the functionality and engagement of fitness tracking platforms.

The project demonstrates the use of:

• Python for data cleaning and preprocessing
• SQL for database management and querying
• Power BI for interactive dashboards and visualization

2️⃣ Business Task

Strava collects large amounts of fitness data from smart devices, but the company does not clearly understand user behaviour patterns. The business objective is to analyze fitness tracker data to understand:

• How users interact with fitness tracking devices
• Whether users maintain healthy activity levels
• How activity impacts calories burned
• How sleep and weight relate to physical activity
• Are users physically active?
• Are users sleeping enough?
• Are users spending too much time sedentary?
• Does higher activity lead to better health outcomes?

Without these insights, Strava cannot design effective marketing campaigns or engagement strategies. These insights help fitness applications improve **user engagement strategies and personalized health recommendations**.

3️⃣ Data Sources

The dataset used for this analysis contains fitness tracker data collected from wearable devices.

The data includes multiple tables representing different aspects of user health and activity.

Dataset	Description
daily_activity	Daily steps, calories, and activity minutes
sleep_day	Sleep duration and time spent in bed
heartrate_daily	Heart rate measurements
hourly_intensity	Hourly activity intensity
weight_log	Weight, BMI, and body fat percentage

4️⃣ Tools Used

The analysis was conducted using the following tools:

Tool	Purpose
Python	Data cleaning and preprocessing
SQL	Data transformation and analysis
MySQL Workbench	Database management
Microsoft Power BI	Data visualization and dashboards
Pandas	Data manipulation
Matplotlib	Exploratory visualization

5️⃣ Data Cleaning and Preparation

Before performing the analysis, the datasets were cleaned and transformed to ensure data quality.

Data Cleaning Steps

The following cleaning steps were applied:

• Removed duplicate records
• Standardized date and time formats
• Filtered unrealistic heart rate values
• Aggregated high-frequency datasets into daily metrics
• Joined datasets into a single master table
✓ Check missing values and Convert date columns

Python Cleaning Example

Heart rate data contained extreme or unrealistic values. These were filtered using Python.

```
heartrate = heartrate[heartrate['Value'] > 30]

heartrate = heartrate[heartrate['Value'] < 220]
```

This ensured that only realistic human heart rate values were included.

6️⃣ Data Transformation in SQL

Due to the large size of the heart rate dataset, the data was summarized into daily statistics.

Daily Heart Rate Summary

```
CREATE TABLE heartrate_summary AS
SELECT
   Id,
   DATE(Time) AS date,
   AVG(Value) AS avg_heartrate,
   MAX(Value) AS max_heartrate,
   MIN(Value) AS min_heartrate
FROM heartrate_daily
GROUP BY Id, DATE(Time);
```

This transformation significantly reduced dataset size and improved analysis performance.

7️⃣ Creating the Master Analytical Dataset

To simplify dashboard creation, all datasets were merged into a single table.

```
CREATE TABLE fitness_master AS
SELECT
a.Id,
a.ActivityDate,
a.TotalSteps,
a.TotalDistance,
```

```sql
a.Calories,
a.VeryActiveMinutes,
a.FairlyActiveMinutes,
a.LightlyActiveMinutes,
a.SedentaryMinutes,
s.TotalMinutesAsleep,
s.TotalTimeInBed,
h.avg_heartrate,
h.max_heartrate,
h.min_heartrate,
w.WeightKg,
w.BMI,
w.Fat
FROM daily_activity a
LEFT JOIN sleep_day s
ON a.Id = s.Id
AND DATE(a.ActivityDate) = DATE(s.SleepDay)
LEFT JOIN heartrate_summary h
ON a.Id = h.Id
AND a.ActivityDate = h.date
LEFT JOIN weight_log w
ON a.Id = w.Id
AND DATE(a.ActivityDate) = DATE(w.Date);
```

- **This master dataset was used for building the Power BI dashboards.**

8️⃣ Exploratory Data Analysis

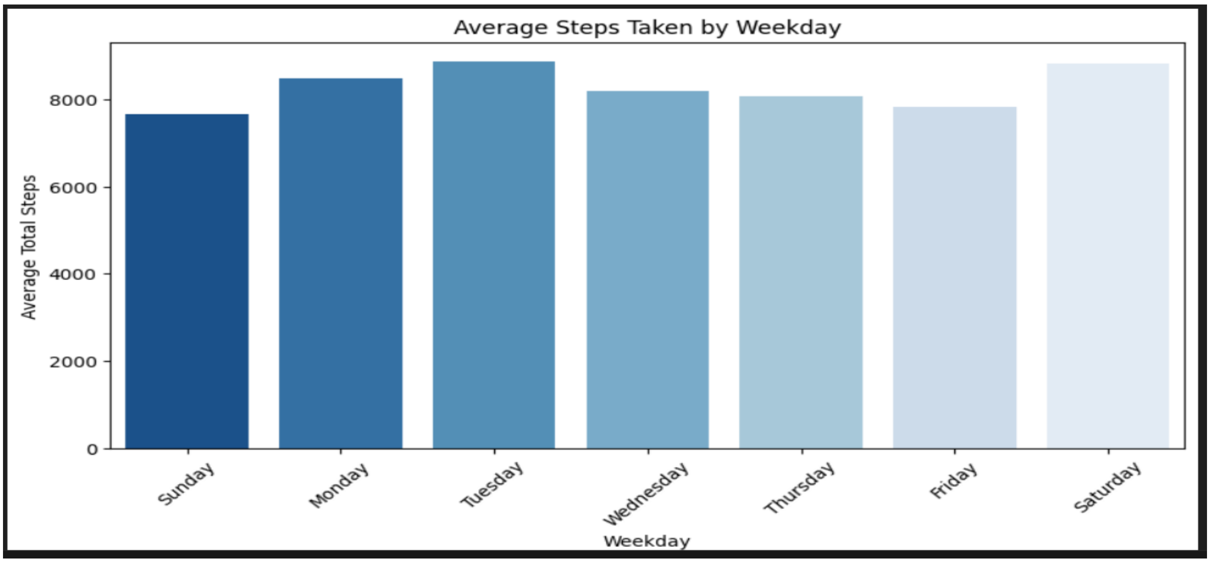
The exploratory analysis focused on three major areas:

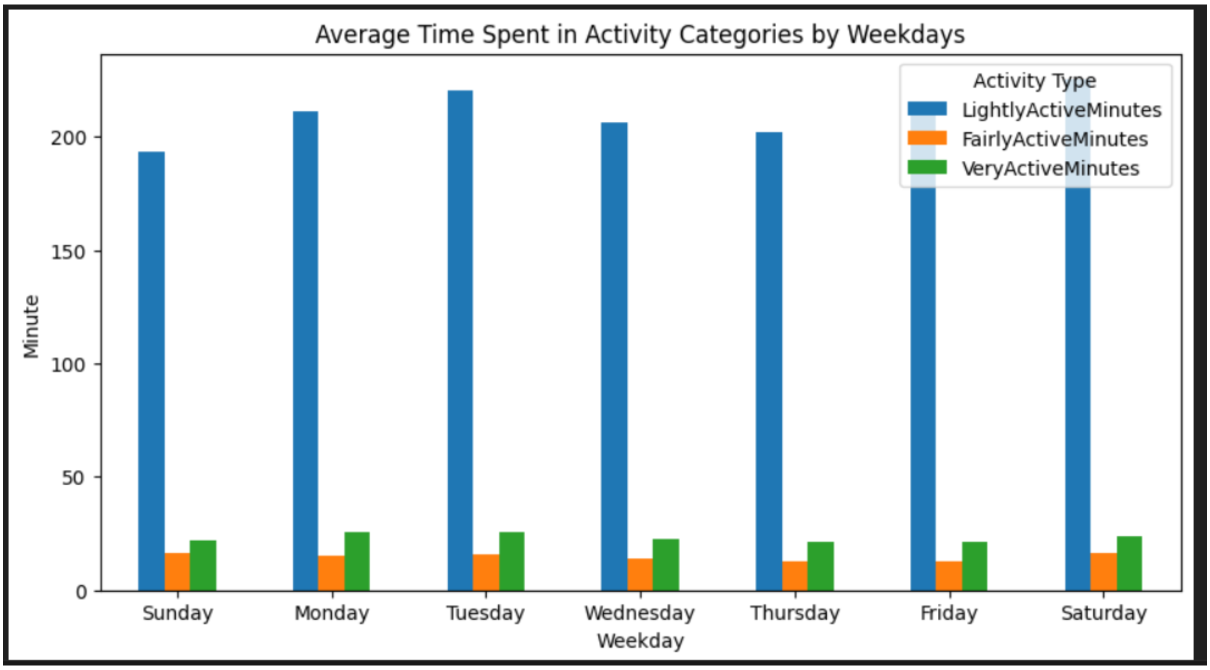
User Engagement

Understanding how frequently users interact with the fitness tracker.

Metrics analyzed:

• Daily steps
• Active minutes
• Hourly activity intensity

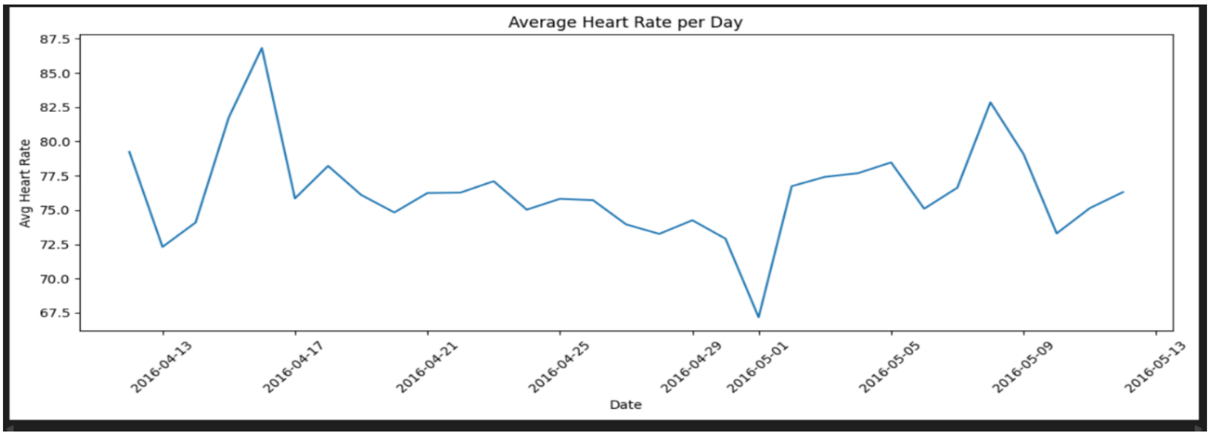


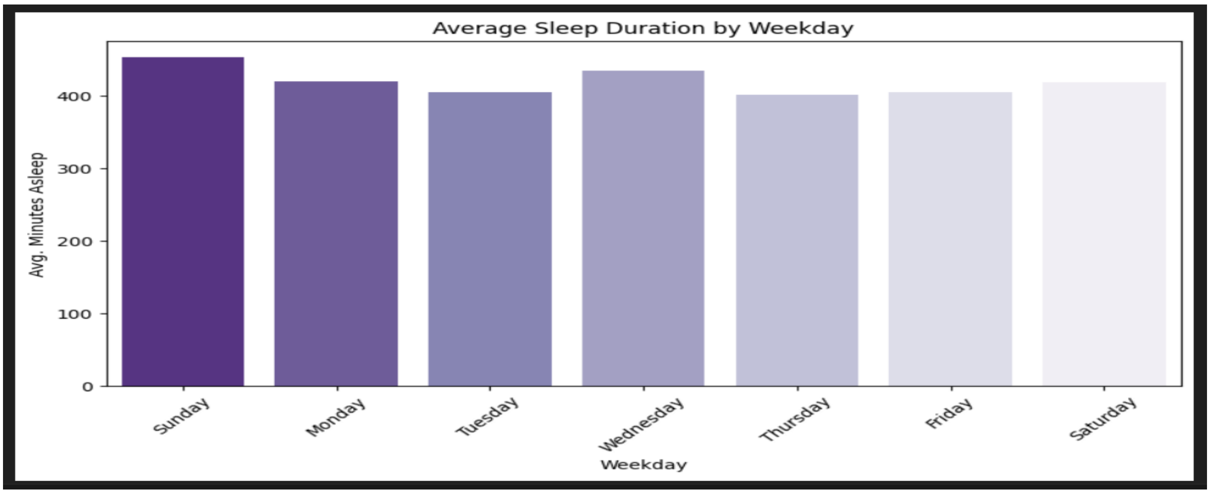


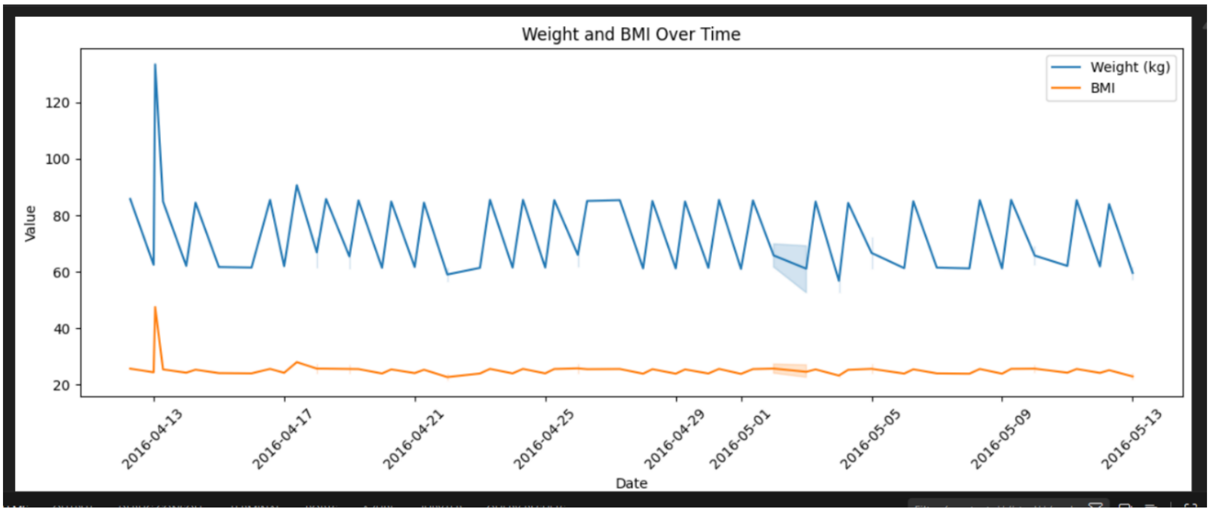
Health Indicators

Evaluating health-related metrics such as:

• Average heart rate
• Sleep duration
• Body Mass Index (BMI)





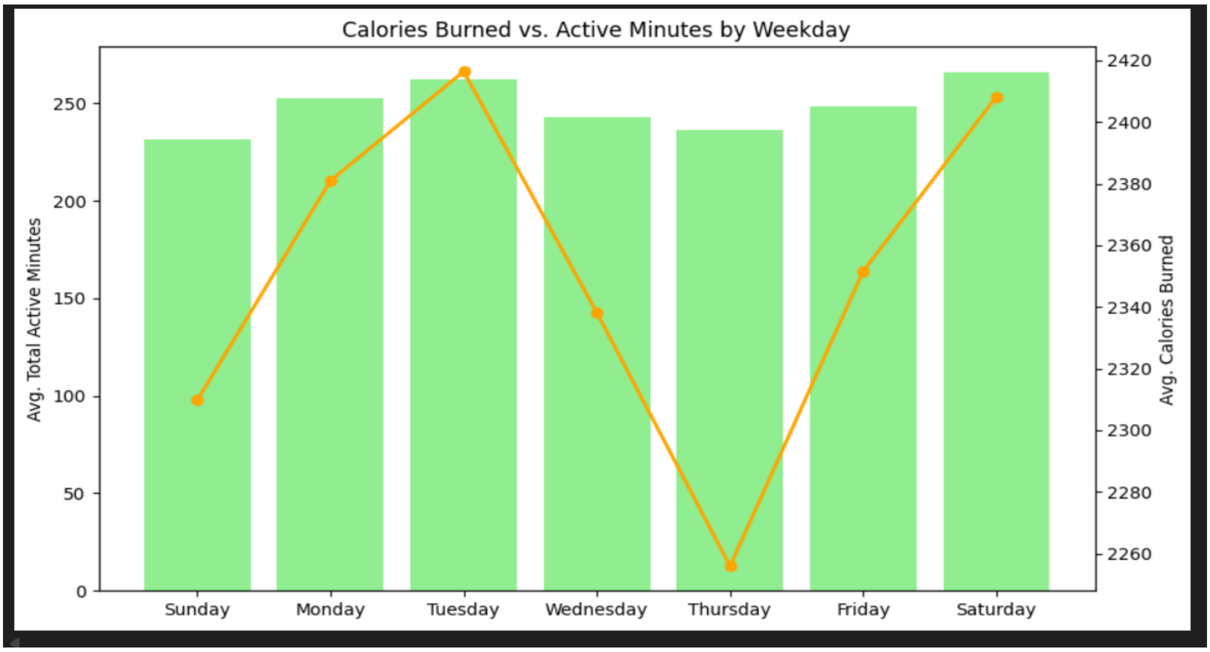


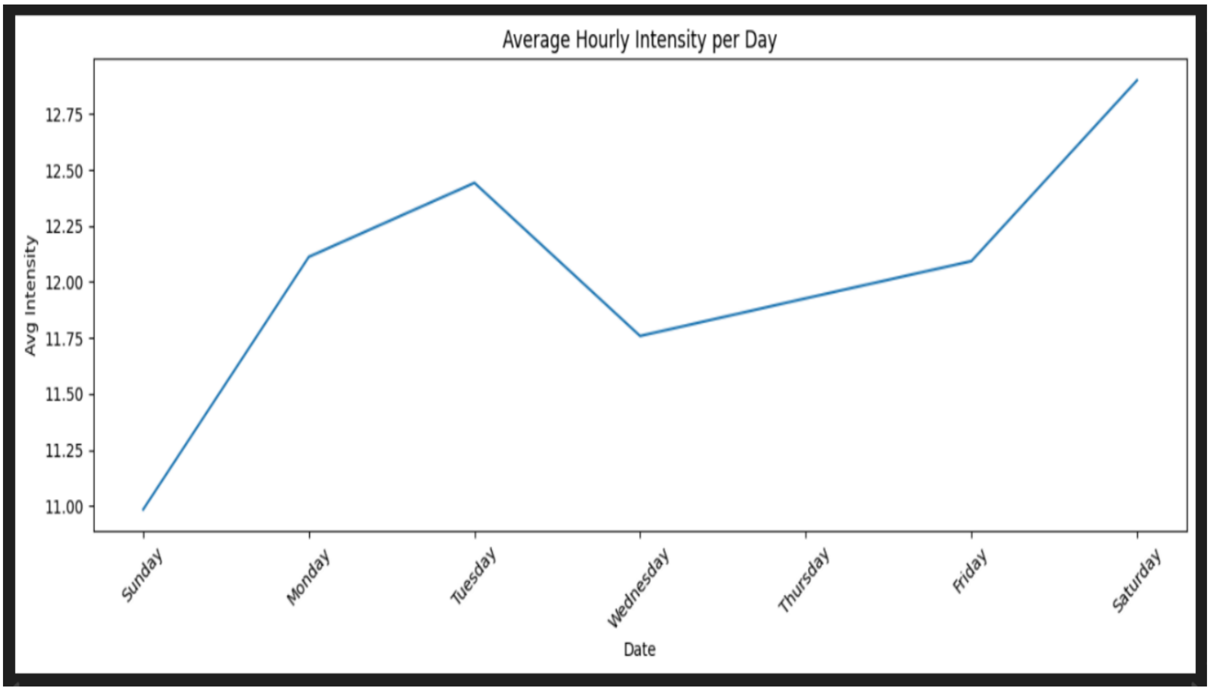
Physical Activity Performance

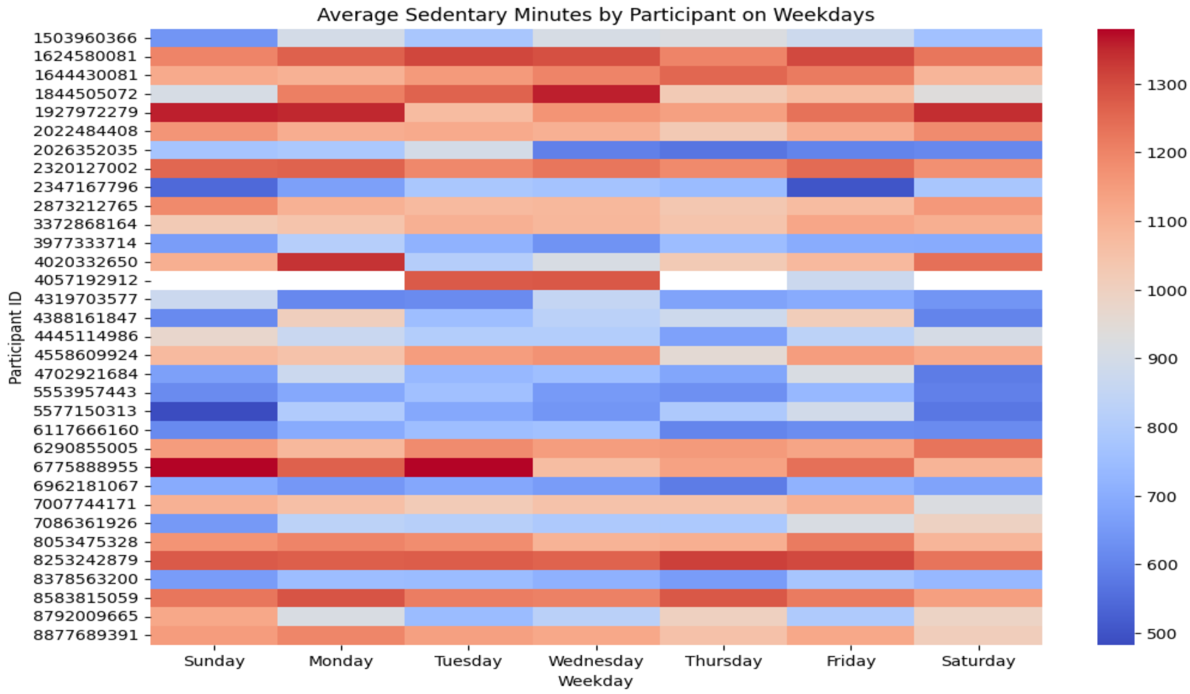
Analyzing how user activity translates into calorie expenditure.

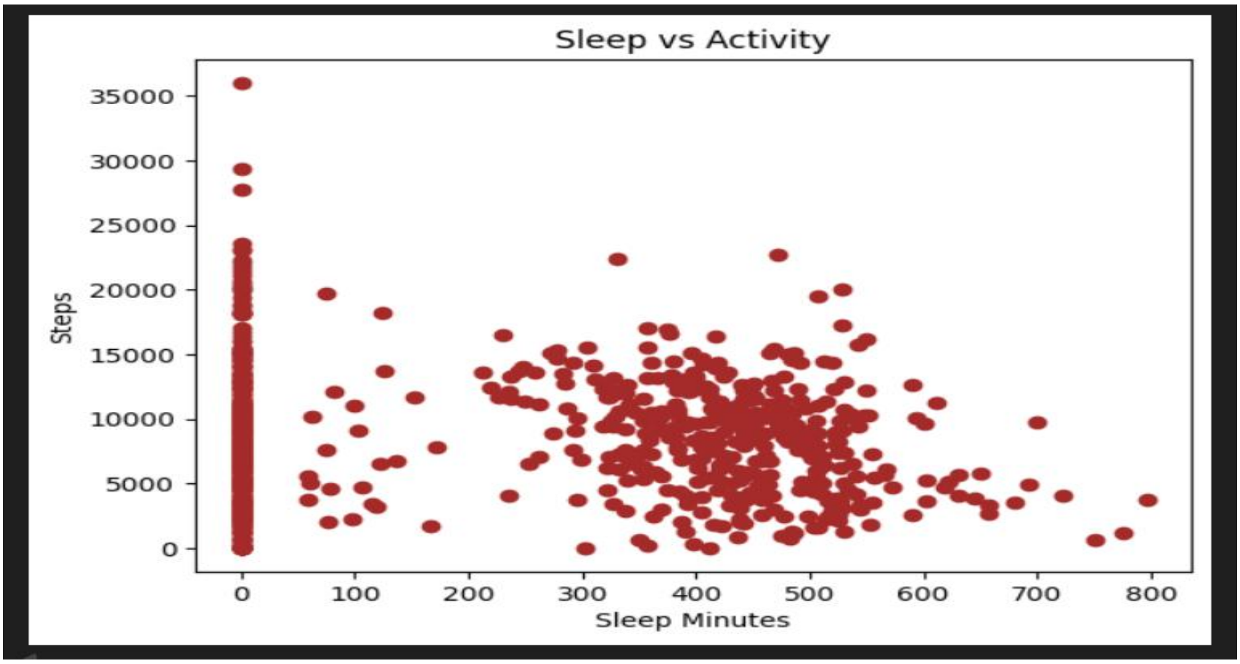
Metrics:

- Calories burned
- Activity intensity
- Sedentary vs active time









9️⃣ Data Visualization

<u>SQL Analysis (Ad-Hoc Queries)</u>

Some SQL Queries -

- ➢ Average Steps per User
- ➢ Average Calories Burned
- ➢ Average Sleep Duration
- ➢ Most Active Users
- ➢ Sedentary vs Active Minutes

- Activity vs Sleep Relationship
- BMI Categories

```
SELECT
        CASE
         WHEN BMI < 18.5 THEN 'Underweight'
         WHEN BMI BETWEEN 18.5 AND 24.9 THEN 'Normal'
         WHEN BMI BETWEEN 25 AND 29.9 THEN 'Overweight'
         ELSE 'Obese'
         END AS BMI_Category,
         COUNT(*)
         FROM weight_log
         GROUP BY BMI_Category;
```

- Activity Consistency

  Some users are active only on certain days.

```
SELECT
Id,
COUNT(DISTINCT ActivityDate) AS Active_Days
FROM daily_activity
GROUP BY Id
ORDER BY Active_Days DESC;
```

  This shows **which users consistently track activity**. Apps like **Strava** use this to measure **user retention**.

- Most Active Users (Leaderboard)

```
SELECT
         Id,
         SUM(TotalSteps) AS Total_Steps
         FROM daily_activity
         GROUP BY Id
         ORDER BY Total_Steps DESC
         LIMIT 10;
```

<u>Three dashboards were created in Power BI -</u>

Dashboard 1 — User Engagement

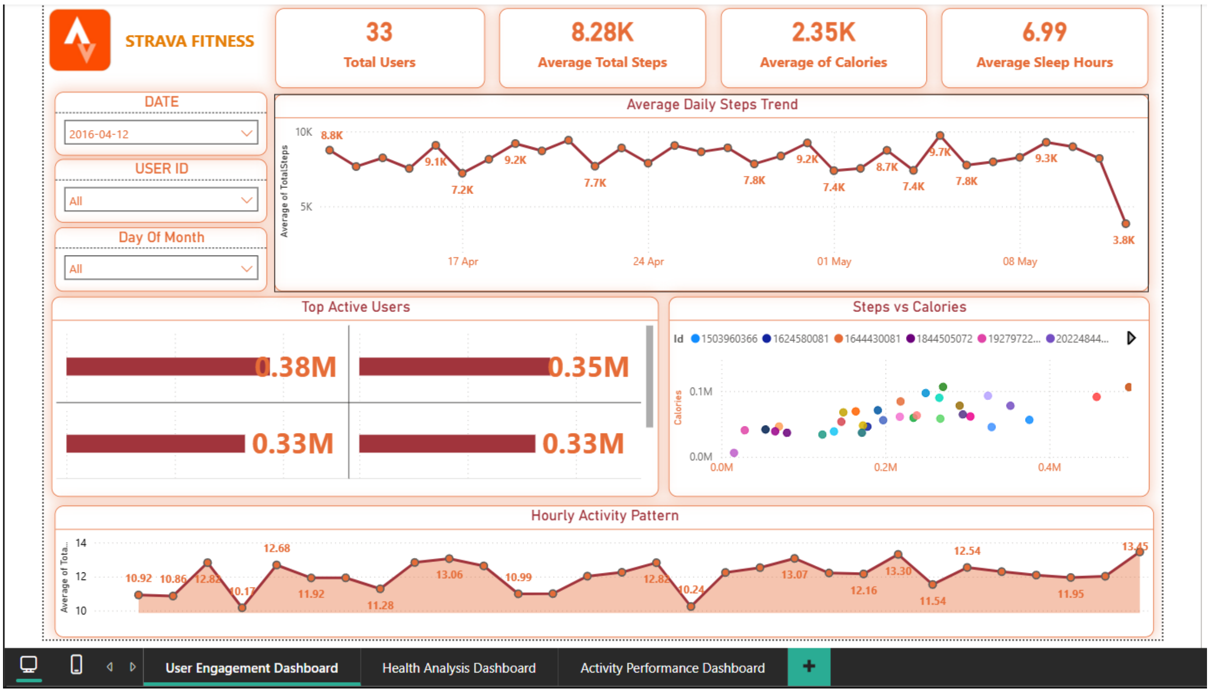
KPI's – Total Users, Avg. Total Steps, Avg. Calories and Average Sleep Hours

Slicers – Date (Year/Month/Day), Users Id and Day of the Month

Visualizations:

- Daily Steps Trend
- Top Active Users
- Steps vs Calories Scatter Plot
- Hourly Activity Distribution

Purpose:
Understand how frequently users engage with the fitness tracker.



Dashboard 2 — Health Analysis

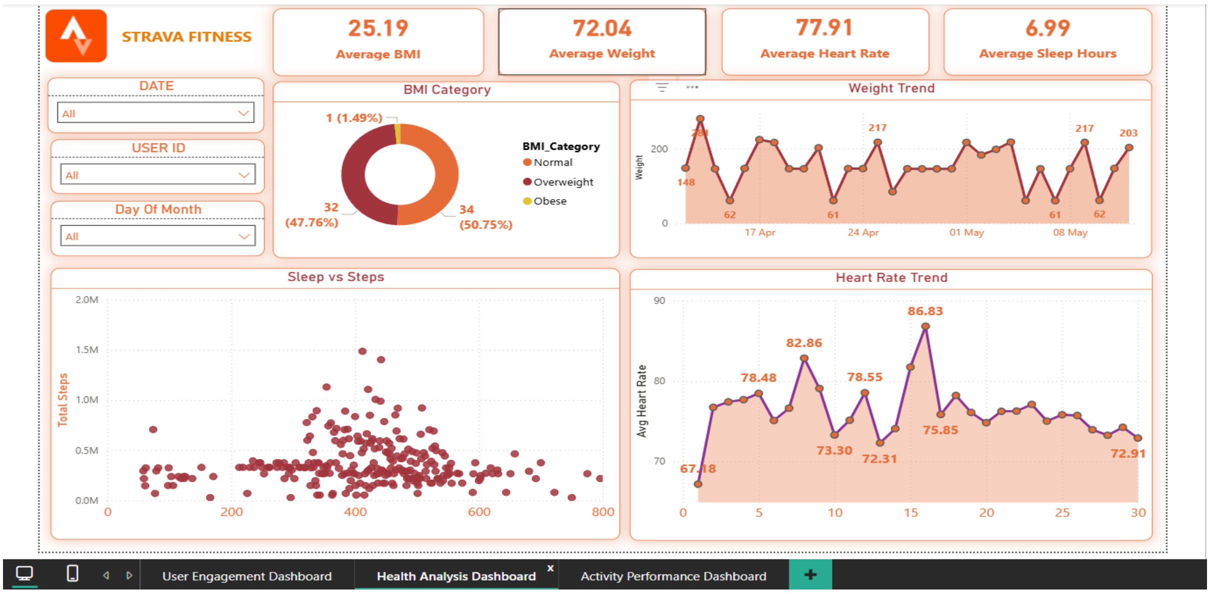
KPI's – Avg. BMI, Avg. Weight, Avg. Heart Rate and Average Sleep Hours

Slicers – Date (Year/Month/Day), Users Id and Day of the Month

Visualizations:

- BMI Category Distribution
- Heart Rate Trend
- Sleep vs Activity Scatter Plot
- Weight Trend Over Time

Purpose:
Evaluate the health indicators of users.



Dashboard 3 — Activity Performance

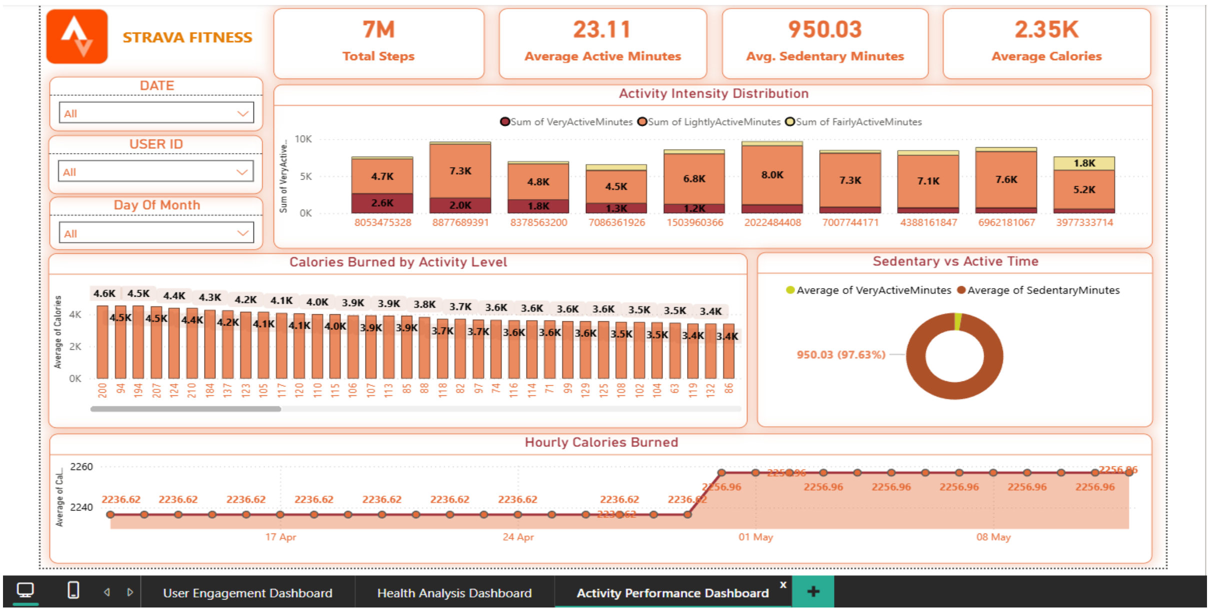
KPI's – Total Steps, Avg. Active Minutes, Avg. Sedentary Min. and Avg. Calories

Slicers – Date (Year/Month/Day), Users Id and Day of the Month

Visualizations:

- Calories Burned Distribution
- Activity Intensity Breakdown
- Sedentary vs Active Time
- Hourly Calories Burned

Purpose:
Measure workout performance and physical activity patterns.



🔟 Business Questions & Answers

1️⃣ What is the average number of daily steps taken by users?
Answer: The average number of steps taken by users per day helps measure overall activity levels. If the average is significantly below the recommended 10,000 steps, it indicates that many users are not reaching optimal activity levels.

2️⃣ How many users reach the recommended 10,000 steps per day?
Answer: This shows how frequently users achieve recommended activity levels. A low count suggests that many users may benefit from fitness challenges or activity reminders.

3️⃣ What is the relationship between steps taken and calories burned?
Answer: This analysis helps identify whether higher physical activity leads to increased calorie burn. Typically, users who take more
steps tend to burn more calories.

4️⃣ Which users are the most active?
Answer: This query identifies the most active users based on total step count. These users may represent highly engaged users of the fitness application.

5️⃣ How much time do users spend in different activity levels?
Answer: This analysis helps understand how users distribute their daily time between active and sedentary behaviors.

6️⃣ What is the average sleep duration of users?
Answer: Sleep duration helps determine whether users are meeting the recommended 7–8 hours of sleep per night.

7️⃣ What is the distribution of BMI categories among users?
Answer: This analysis shows the health distribution of users based on BMI categories.

8️⃣ What is the average heart rate of users?
Answer: Average heart rate provides insights into users' cardiovascular health and exercise intensity.

9️⃣ When are users most active during the day?
Answer: This identifies the time of day when users are most physically active.

🔟 What percentage of time do users spend sedentary?
Answer: This analysis highlights how much time users spend inactive during the day

1️⃣1️⃣ Key Insights

Insight 1 — Many users do not reach the recommended step count

The average number of daily steps was lower than the commonly recommended 10,000 steps per day, suggesting moderate user activity levels.

Insight 2 — Users are most active in the evening

Activity intensity peaks during late afternoon and evening hours, indicating that users exercise primarily after work or school hours.

Insight 3 — Strong relationship between steps and calories burned

Users with higher step counts consistently burned more calories, confirming the direct relationship between physical activity and energy expenditure.

Insight 4 — High sedentary time

Many users spend a significant portion of the day in sedentary behavior, indicating potential opportunities to promote more frequent movement.

Insight 5 — Sleep patterns influence activity levels

Users with longer sleep durations generally show higher daily activity levels.

Insight 6 — BMI distribution indicates mixed health levels

BMI analysis shows that while many users fall within the normal range, a portion of users fall into the overweight category.

Insight 7 — Many users do not reach recommended daily step goals

Analysis of the daily activity data shows that a large number of users do not achieve the commonly recommended 10,000 steps per day. This indicates that many users maintain only moderate levels of physical activity.

Insight 8 — High sedentary time among users

The dataset shows that users spend a significant portion of their day in sedentary minutes, indicating long periods of inactivity. Prolonged sedentary behavior may negatively impact overall health and fitness levels.

Insight 9 — Some users do not meet recommended sleep duration

Sleep analysis indicates that several users sleep less than the recommended 7–8 hours per night. Insufficient sleep can impact energy levels, recovery, and physical performance.

Insight 10 — Higher activity leads to higher calorie burn

The analysis shows a strong positive relationship between daily steps and calories burned. Users with higher activity levels tend to burn more calories, confirming the expected relationship between physical activity and energy expenditure.

Insight 11 — Some users fall into the overweight BMI category

The BMI distribution analysis shows that while many users fall into the normal BMI range, a portion of users are classified as overweight. This indicates opportunities for fitness applications to provide targeted health recommendations.

1️⃣2️⃣ Business Recommendations

Based on the analysis, the following recommendations can help improve user engagement and health outcomes.

Introduce Activity Challenges

Gamified step challenges can motivate users to reach daily activity goals.

Encourage Regular Movement

The application could implement inactivity alerts to reduce sedentary behaviour.

Personalized Health Insights

Using BMI, heart rate, and sleep data, the platform can generate personalized fitness recommendations.

Optimize Engagement Timing

Notifications and workout suggestions should be scheduled during peak activity hours.

Introduce Personalized Fitness Programs

Using user data such as daily steps, heart rate, and BMI, the application could generate personalized workout plans tailored to individual fitness levels and goals. This would help users gradually improve their health and maintain motivation.

Provide Sleep Improvement Suggestions

Since sleep patterns influence physical activity levels, the platform could offer sleep recommendations and bedtime reminders. Features such as sleep quality insights and recovery tracking could help users maintain healthier routines.

Gamify User Engagement

Introducing leaderboards, badges, and fitness challenges can encourage users to stay active and compete with friends or community members. Gamification can significantly increase daily user engagement.

Activity Goal Tracking

The application could allow users to set custom fitness goals, such as daily step targets, calorie burn goals, or workout duration goals. Progress tracking and milestone notifications can motivate users to maintain consistency.

Smart Notifications and Reminders

Based on the analysis of peak activity hours, the platform can send personalized reminders during times when users are most likely to be active, such as evening workout prompts or morning activity reminders.

Reduce Sedentary Behavior

To address long sedentary periods, the app could implement movement reminders or short activity suggestions when users remain inactive for extended durations.

Health Risk Awareness

For users classified as overweight based on BMI, the application could provide health awareness notifications and lifestyle recommendations to encourage healthier habits.

Integrate Social and Community Features

Allowing users to share achievements, join fitness groups, and participate in community challenges can create a sense of motivation and accountability.

AI-Based Health Insights

By analyzing long-term fitness data, the platform could provide AI-driven health insights, predicting potential health risks and recommending lifestyle improvements.

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1 3 Conclusion

This analysis highlights how wearable fitness data can reveal meaningful insights about user behaviour, health patterns, and physical activity trends.

By leveraging these insights, fitness platforms such as Strava can enhance user engagement and promote healthier lifestyles through personalized and data-driven features.

Future Improvements and Scope

While this analysis provides valuable insights into user activity and health patterns, there are several opportunities to expand the project and improve the depth of analysis.

Expand the Dataset

The current dataset contains activity data from a limited number of users and time periods. Future analysis could include a **larger dataset with more users and longer time ranges** to generate more reliable insights and identify long-term fitness trends.

Include Additional Health Metrics

Additional health indicators such as **hydration levels, nutrition data, stress levels, and workout types** could provide a more comprehensive understanding of user health and lifestyle patterns.

These features are commonly integrated into modern fitness platforms such as **Strava** and **Fitbit**.

Apply Predictive Analytics

Future work could include **machine learning models** to predict user behavior, such as:

• Predicting whether users will reach their daily step goals
• Forecasting calorie burn based on activity levels
• Identifying users at risk of inactivity or poor health indicators

Predictive analytics could help fitness applications provide **proactive health recommendations**.

Real-Time Data Monitoring

The current analysis uses historical data. Future systems could integrate **real-time data streaming from wearable devices** to provide instant insights, live dashboards, and dynamic health alerts.

Advanced User Segmentation

Users could be segmented into categories such as:

• Highly active users
• Moderately active users
• Sedentary users

This segmentation could help fitness applications deliver **personalized experiences and targeted recommendations**.

Integration with Mobile Dashboards

Future work could involve deploying the dashboards into **mobile-friendly analytics applications** so users and health professionals can easily monitor health metrics on the go.

Final Project Outcome

This project demonstrates how fitness tracker data can be analyzed to uncover patterns in **physical activity, sleep behavior, and health indicators**.

By applying **data cleaning, SQL analysis, and visualization techniques**, meaningful insights were generated that can help fitness platforms improve **user engagement and health outcomes**.

The dashboards created using **Microsoft Power BI** provide an interactive way to explore user fitness behavior and support data-driven decision-making.

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